Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/14/2024
PI Name:	Shelhamer, Mark Sc.D.		
Project Title:	Assessment of Otolith Function and Asymmetry as a Corollary to Critical Sensorimotor Performance in Missions of Various Durations		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mshelhamer@jhu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	410-614-5898
Organization Name:	Johns Hopkins University		
PI Address 1:	Otolaryngology - Head and Neck Surgery Department		
PI Address 2:	710 Ross Bldg, 733 N. Broadway		
PI Web Page:	https://		
City:	Baltimore	**State:**	MD
Zip Code:	21205-1832	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	02/07/2019	**End Date:**	02/07/2033
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/7/2033 per L. Barnes-Moten/JSC and NSSC information (Ed., 5/18/21) NOTE: End date changed to 6/6/2021 per NSSC information (Ed., 11/9/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schubert, Michael Ph.D. (Johns Hopkins University) Kheradmand, Amir M.D. (Johns Hopkins University School of Medicine)		
Grant/Contract No.:	80NSSC19K0487		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose a simple and rapid measure of otolith asymmetry -- a basic low-level neurovestibular function that is altered in spaceflight. Our assessment apparatus and procedure was developed with previous Human Research Program (HRP) support, and has undergone validation testing and is currently in use in an unrelated Department of Defense (DoD) study. It provides a measure of binocular alignment (vertical and torsional), which is a reflection of underlying otolith asymmetry. While this measure might be important in itself (since it reflects a form of central neural compensation that might be challenged in spaceflight), it could also provide a simple alternative to locomotion and posture testing if it is found to correlate with those measures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A simple and rapid assessment of neural/neurovestibular function, as described here, will be a useful in-flight procedure to track changes over the course of a mission. Given the widespread connection of neural circuits and functions with other body functions, it is not inconceivable that neurovestibular effects as assessed here will likewise have implications for a wide variety of neural, cognitive, and performance aspects of crews in flight. Our testing provides an opportunity to assess a low-level neural system that is not under volitional control, is responsive to g-level alterations, and might be an indicator of broader and more functionally relevant effects. The technology and procedures for this project are being used in DoD projects for the assessment of vestibular function in military personnel.
Task Progress:	Online integrated consent briefings were supported, and a presentation was made at the NASA Human Research Program (HRP) Investigators Workshop in February 2024. In the fall of 2021, we started a set of experiments in the lab to provide further baseline data by which to evaluate the upcoming flight data from this project. In these trials, subjects are evaluated for ocular alignment (OA) while standing on a hard floor, standing on foam, while seated, and while making head movements. This will help us evaluate the impact of postural demands on ocular alignment; we conjecture that the heightened demand while standing will call for a more precise compensation for any latent otolith asymmetry, causing a reduction in misalignment in that condition compared to seated. A new student team is now working on this project, Institutional Review Board (IRB) approvals and training have been completed, and experiments resumed in 2024. At this time approximately 25 subjects have been tested, and data analysis is continuing. A portion of the protocol for this project was modified and extended for use as an educational demonstration at Space Center Houston. This enables collection of a large set of normative data (N=167 currently) with which to refine our analyses, and which can provide context for interpretation of results from astronauts. Initial analysis of these data were performed by medical student Nabila Ali for an epidemiology degree project and is now being prepared for publication. In parallel, biostatistics graduate student Alyssa Columbus is re-analyzing these data to look for relationships between the posture findings and the ocular-alignment (otolith asymmetry) findings. This project also includes a component dedicated to data-integration of the other CIPHER (Complement of Integrated Protocols for Human Research) projects. While awaiting CIPHER flight data, we have been performing several preparatory activities so that we can be better prepared to organize and analyze that data: 1. An updated set of hypotheses (e.g., expected cross-study associations or correlations) was prepared, per the Statement of Work for this project, and provided to HRP in March 2024. In summary, some of the hypothesized cross-discipline connections include: •Vestibular and Cognitive. Disruption of the vestibular contribution to spatial orientation and memory might be a source of “space fog” and might be reflected in parallel changes in vestibular-mediated functions and cognitive tests. •Vestibulo-Autonomic interactions. These interactions impact post-flight orthostatic intolerance. Correlating inflight and postflight vestibular measures with postflight reports of presyncope or stumbling would provide information on this connection. •Bone and Vestibular. The vestibular gravity sensors rely on calcium carbonate, and calcium is the predominant constituent of bone. Thus, there might be parallel changes in ocular alignment (mediated by the otolith organs) and changes in bone structure. Connection to nutrition and food is also likely, as a source of calcium. •Fluid-shift effects related to the Eye and Sleep (diurnal phenomena). Sleep is essential in clearing neural waste products, which might be adversely impacted by alterations in fluid flow consequent to cephalid fluid shifting. Thus, there may be correlations between sleep quality and amount, ocular and other measures of fluid shift, and cognitive function dependent on neural integrity. •Stress markers and Cognition, Sleep, and Performance. It might not be possible to identify markers of stress that are not also indicators of other effects, but it might be possible to create a useful surrogate marker for stress that incorporates correlated measures of (for example) heart rate variability (HRV), sleep disturbance, cortisol, and other markers related to stress. Then, this stress indicator can be probed for correlations with other functions such as cognition and performance. Performance markers could include errors in routine tasks and calls to MCC for assistance. •Stress and Cardiac and Immune function. As noted, stress per se can be difficult to assess, but similar reasoning might be applied to look for correlations between stress and cardiovascular measures, and stress and immune function. •Vestibular and Teamwork and other Social interactions. Vestibular disorders can have an impact on motivation and interpersonal relations, which might impact behavioral and interpersonal issues such as relationships with others and the environment. •Circadian rhythms, Body temperature, and Sleep. Circadian rhythms would be derived from prevailing sleep cycle, and disruption inferred from alarms or slam shifts (metadata). Body temperature might not be available unless there is a surrogate measure such as a verbal report. •Sleep and Consolidation of learning/memory. Memory and motor learning are consolidated during sleep. It might be possible to determine if there are connections between sleep alterations and problems with memory or execution of motor skills. •Stress and Nutrition and Exercise. Nutrition and exercise countermeasures for physiological deconditioning, and are relatively easy to assess (biochemical profile, exercise logs). Stress is more difficult to assess and is multifactorial. Nevertheless, nutrition and exercise are known to have large effects on stress coping and so some effort should go into determining if there are measurable relationships. 2. A manuscript describing the scientific background and justification for CIPHER has been prepared by the PI. The final editing of this manuscript is still taking place. 3. Discussions began on how to acquire, archive, and curate the data that are expected to be provided to the PI by the other CIPHER investigators. After talking with several groups, it was determined that NLSP would be the right platform, provided a means for incorporating medical and metadata is developed, and the PI can have access to a work area on the platform. 4. Other discussions with ALSDA (Ames Life Sciences Data Archive) took place to identify datasets that are already

	available, that could help address some of the same questions as CIPHER. 5. Through discussions with Anand Narayanan, we have identified a set of data in OSDR (Rodent Research 9) that can be analyzed for relations between bone and ocular measures, as an initial investigation of cross-disciplinary effects of spaceflight. Since this project will analyze significant data before CIPHER, it can help provide guidance as to some relevant cross-discipline analysis methods. This is assisted by data science student Xiangyu Zhang and her faculty advisor Brian Caffo.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Alwood JS, Mulavara AP, Iyer J, Mhatre SD, Rosi S, Shelhamer M, Davis C, Jones CW, Mao XW, Desai RI, Whitmire AM. "Circuits and biomarkers of the central nervous system relating to astronaut performance: summary report for a NASA-sponsored technical interchange meeting" Life (Basel). 2023 Aug 31;13(9)1852. https://doi.org/10.3390/life13091852 ; PubMed PMID: 37763256; PubMed Central PMCID: PMC10532466 , Aug-2023
Articles in Peer-reviewed Journals	Reynolds RJ, Shelhamer M, Antonsen EL, Carpentier WR. "Characterizing dehydration in short-term spaceflight using evidence from Project Mercury." npj Microgravity. 2024 Jun 11;10:64. https://doi.org/10.1038/s41526-024-00374-8 ; PubMed PMID: 38862554; PubMed Central PMCID: PMC11166991 , Jun-2024
Articles in Peer-reviewed Journals	Chiaberge M, Thottappillil N, Liphardt A-M, Furlanetto A, Odell D, Wang C, Hope S, Smee S, Rehfus J, Niehoff A, Shelhamer M, Norman C, Philippon MJ, Huard J, James AW, Fan C-M. "Plyometric training increases thickness and volume of knee articular cartilage in mice." npj Microgravity. 2025 Feb 13;11:5. https://doi.org/10.1038/s41526-025-00458-z ; PMID: 39948108; PMCID: PMC11825961 , Feb-2025
Books/Book Chapters	Shelhamer M. "Preparing for the unpredictable: Facilitating multisystem resilience in human spaceflight." in "Building a Space-Faring Civilization." Ed. M.A. Schmidt, M.J. Legato. Cambridge, MA: Academic Press, 2025. p. 195-203. https://doi.org/10.1016/B978-0-443-13850-8.00023-3 , Feb-2025
Books/Book Chapters	Shelhamer M, Antonsen E. "Systems Medicine in Human Spaceflight." Ed. M. Shelhamer, E. Antonsen. New Jersey: World Scientific, 2024. http://dx.doi.org/10.1142/13713 , May-2024
Books/Book Chapters	Shelhamer M, Fogarty J. "Crew health – Psychological, biological, and medical issues and the need for a systems approach." in "Interstellar Travel: Propulsion, Life Support, Communications, and the Long Journey" Ed. L. Johnson, K. Roy. Elsevier, 2024. p. 81-114. https://doi.org/10.1016/C2021-0-01074-9 , May-2024
Books/Book Chapters	Shelhamer M, Reynolds RJ, Antonsen E "Predictive Modeling of Spaceflight Medical Issues" in "Systems Medicine For Human Spaceflight" Ed. M Shelhamer, E Antonsen World Scientific, May-2024
Papers from Meeting Proceedings	Shelhamer M, Gersh JR. "A Mission Architecture to Integrate Human and Spacecraft Functional Performance." 2024 IEEE Aerospace Conference, Big Sky, Montana, March 2-9, 2024. 2024 IEEE Aerospace Conference, Big Sky, Montana, March 2-9, 2024. , Mar-2024
Papers from Meeting Proceedings	Shelhamer M, Antonsen E, Reynolds R. "The Need for a Systems Approach in Spaceflight Health and Medicine." 2025 IEEE Aerospace Conference, Big Sky, Montana, March 1-8, 2025. 2025 IEEE Aerospace Conference, Big Sky, Montana, March 1-8, 2025. IEEE Aerospace Conference , Mar-2025